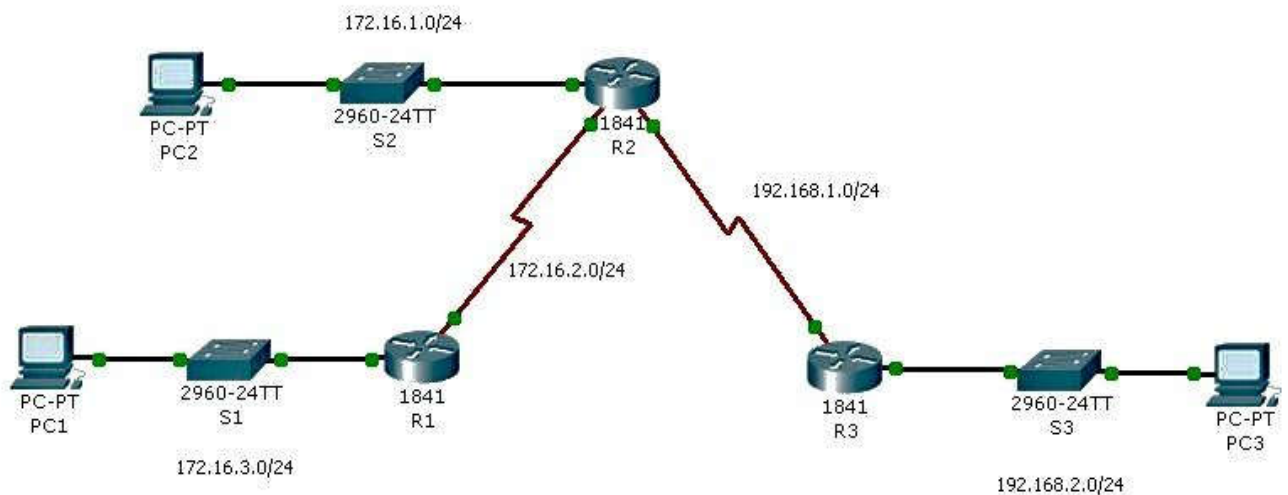


The Computer Networks

Lab 4

TP Static Routing - 2

The purpose of this Lab is to implement a static routing configuration.



3. Configure serial interfaces and verification of changes to the routing table

The goal is to apply the configuration of serial interfaces. We will also use debug ip routing to analyze the update process of the routing table.

Presentation:

Routers connect to routers on other sites that use different WAN technologies. One technology is to rent a dedicated line from a service provider and connect to its network via the CSU/DSU unit connected to a serial interface on the router. The CSU/DSU is a data communication equipment (DCE), which provides a clock signal for the circuit. In this lab, we will use a crossover cable to directly connect two routers via their serial ports. A clock signal is transmitted from the router connected to the DCE end of the cable. This exercise describes how to configure the IP information and a clock signal to the router serial interfaces.

While the interfaces are configured for IP, the directly connected networks are added to the IP routing table. It is this process which we will examine in the context of this exercise.

Educational goals:

- Configure serial interfaces on routers
- Configure IP information and timing on the router serial interfaces
- Check the status of interfaces on routers
- Check the information in the routing table
- Check the information in the routing table for directly connected networks.
- Observe the process of updates to the routing table

Task 1: Configuration of serial interfaces on routers

Step 1: Set the IP information and timing on the router serial interfaces.

Go to the router R1. Under the CLI tab, enter the privileged EXEC mode by entering the enable command. Enter the global configuration mode by entering the command **conf t**. Enter the configuration mode for the first serial interface by entering the command **interface s0/0/0**. Configure the IP address with the command **ip address 172.16.2.1 255.255.255.0**. The DCE end of the cable is connected to this interface. Set up synchronization by entering the clock rate command **64000**. Activate the interface by entering the **no shutdown** command. Exit configuration mode by using the keyboard shortcut **Ctrl + z**. Save the configuration by running the command **copy run start**. Repeat these steps for the other two routers, with the following information.

Peripheral	Interface	IP adress	Subnet mask	Default Gateway
R1	Fa0/0	172.16.3.1	255.255.255.0	N / D
	S0/0/0	172.16.2.1	255.255.255.0	N / D
R2	Fa0/0	172.16.1.1	255.255.255.0	N / D
	S0/0/0	172.16.2.2	255.255.255.0	N / D
	S0/0/1	192.168.1.2	255.255.255.0	N / D
R3	Fa0/0	192.168.2.1	255.255.255.0	N / D
	S0/0/1	192.168.1.1	255.255.255.0	N / D
PC1	NIC	172.16.3.10	255.255.255.0	172.16.3.1
PC2	NIC	172.16.1.10	255.255.255.0	172.16.1.1

PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
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Step 2: Check the status of interfaces on routers.

Check the status of serial interfaces by entering the **show ip interface brief** on the three routers.

Task 2: verification of information on the routing table

Step 1: Check the information about the routing table for directly connected networks.

Check the contents of the IP routing table by entering the **show ip route** command on all three routers. The IP routing table shows directly connected networks and the associated interface.

Step 2: Observe the process of updates to the routing table.

Go to the router R2 and enter interface configuration mode for the interface s0/0/0. Run the **shutdown** command to disable the interface. Do the same with the S0/0/1. Exit configuration mode and enter the **show ip route** command. Networks connected to serial interfaces are no longer in the routing table. Enter the command **debug ip routing**. Activate the two serial interfaces with the **no shutdown** command. Check the entries added to the routing table.

4. Devices on directly connected networks

The objective is to test connectivity between directly connected devices.

Presentation:

Devices connected directly and configured for IP can communicate with each other. Without routing, they cannot communicate with other devices on networks that are not directly connected.

Educational goals:

- Check connectivity between devices
- Verify connectivity between hosts and routers
- Check connectivity between routers

Task 1: Verify connectivity between devices

Step 1: Verify connectivity between hosts and routers.

Run **ping 172.16.3.1** from PC1 prompt commands to perform a ping test of the FastEthernet0/0 interface of router R1. Run **ping 172.16.2.1** from PC1 prompt commands to perform a ping test of the serial0/0/0 of router R1. Both pings succeed because the router R1 is connected directly to PC1. Run **ping 172.16.2.2** since prompt PC1 commands to perform a ping test of the serial0/0/0 of router R2. This ping fails because the router R2 is not connected

directly to PC1. Perform the ping test IP addresses for all routers and interfaces from PC2 and PC3, and note the results.

Step 2: Check the connectivity between the routers.

Run **ping 172.16.3.10** from the CLI on R1 to ping the FastEthernet interface on PC1. Run **ping 172.16.2.2** from the CLI on router R1 to ping the Serial0/0/0 interface on router R2. The two ping requests succeed because the R1 router is directly connected to the PC1 and the Serial0/0 interface on R2 is the same subnet as Serial0/0/0 on the R1 router. Run **ping 172.16.1.1** from the CLI on R1 to ping the FastEthernet0/0 interface on the R2 router. This ping request fails. Indeed, although the router R2 is connected directly to the router R1, the interface FastEthernet0/0 on the router R2 is another subnet. In addition, the IP routing table on R1 router has no route to this destination. You can confirm this by running the **show ip route command** on router R1. Ping IP addresses for all routers and PC interfaces from the R2 and R3 routers, and note the results.